Probability Theory

Learning Objectives

By the end of this session, you will be able to:

- **Define** probability, sample space, and events.
- Understand key properties of probability.
- **Calculate** probabilities for dice, coins, and cards **with step-by-step solutions**.

What is Probability?

- Measures likelihood of an event (Range: 0 to 1).

$$\text{Probability of an event} = \frac{Number\ of\ favourable\ outcomes}{Total\ number\ of\ possible\ outcomes}$$

Basic Probability Concepts

- Probability – the chance that an uncertain event will occur (always between 0 and 1)
- Impossible Event – an event that has no chance of occurring (probability = 0)
- Certain Event – an event that is sure to occur (probability = 1)

Question: What is the probability of rolling a number greater than 4 on a fair six-sided die?



Problem: What is the probability of rolling a number greater than 4 on a fair six-sided die?

Solution: A six-sided die has six possible outcomes: 1, 2, 3, 4, 5, and 6. The favorable outcomes for rolling a number greater than 4 are 5 and 6.

- Number of favorable outcomes: 2 (which are 5 and 6)
- Total number of outcomes: 6

Answer: The probability of rolling a number greater than 4 is 2/6 = 1/3

Key Terms in Probability

Term	Meaning
Experiment	A procedure with uncertain outcome (e.g., rolling dice)
Sample Space (S)	Set of all possible outcomes
Event (E)	One or more outcomes from the sample space
Favorable Outcomes	Outcomes that make the event happen

Reference Dataset

- ‘titanic’ dataset crosstab

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

Simple Calculations

Find out the following probabilities:

1. Probability that a passenger is female (P(Female)).

2. Probability that a passenger is male (P(Male))

3. Probability that a passenger survived (P(Survived))

4. Probability that a passenger is female and survived (P(Female ∩ Survived))

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

Probability that a passenger is female (P(Female))

$$P(Female) = \frac{\text{Total number of females}}{\text{Total passengers}} = \frac{314}{891} \approx 0.3524$$

Probability that a passenger is male (P(Male))

$$P(Male) = \frac{577}{891} \approx 0.6476$$

Probability that a passenger survived (P(Survived))

$$P(Survived) = \frac{342}{891} \approx 0.3838$$

$$P(Female \cap Survived) = \frac{233}{891} \approx 0.2614$$

Properties of Probability

1. Non-negativity

Every probability is a non-negative number, ranging from 0 to 1. This means you cannot have a probability less than 0 or greater than 1.

$$0 \leq P(A) \leq 1$$

for any event A.

2. Normalization

The probability of the sample space S, which represents all possible outcomes of a random experiment, is 1.

$$P(S) = 1$$

Complement Rule

The probability of the complement of an event A (denoted as A^c or $\bar{A}$)—meaning the event that A does not occur—is 1 minus the probability that A does occur.

$$P(A^c) = 1 - P(A)$$

Problem: If the probability of raining tomorrow is 0.3, what is the probability of it not raining?

Probability that a passenger survived (P(Survived))

$$P(Survived) = \frac{342}{891} \approx 0.3838$$

Addition Rule

If two events A and B are not mutually exclusive (meaning they can occur together), the probability that A or B occurs is the sum of their individual probabilities minus the probability of their intersection (the probability that both events occur).

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Problem: What is the probability of drawing a red card or a king from a standard deck of cards?

Problem: What is the probability of drawing a red card or a king from a standard deck of cards?

Solution: $\frac{1}{2} + \frac{1}{13} - \frac{1}{26} = \frac{7}{13}$.

Explanation: Red cards and kings are not mutually exclusive (there are 2 red kings), so subtract the intersection.

Probability that a passenger is female or survived (P(Female ∪ Survived))

Using the **Addition Rule**:

$$P(Female \cup Survived) = P(Female) + P(Survived) - P(Female \cap Survived)$$

Probability that a passenger is female or survived (P(Female ∪ Survived))

$$P(Female \cup Survived) = P(Female) + P(Survived) - P(Female \cap Survived)$$

$$= \frac{314}{891} + \frac{342}{891} - \frac{233}{891} = \frac{423}{891} \approx 0.4748$$

Addition Rule for Mutually Exclusive Events

If two events A and B are mutually exclusive (meaning they cannot both occur at the same time), the probability that A or B occurs is the sum of their individual probabilities.

$$P(A \cup B) = P(A) + P(B)$$

This can be extended to any number of mutually exclusive events.

Problem: What is the probability of rolling a 1 or 2 on a six-sided die?

Problem: What is the probability of rolling a 1 or 2 on a six-sided die?

Solution: $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$.

Explanation: Rolling a 1 and rolling a 2 are mutually exclusive, so you add their probabilities.

Conditional Probability

The probability of an event A given that event B has occurred (denoted $P(A|B)$) is the probability of the intersection of A and B divided by the probability of B, provided that $P(B)$ is not zero.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability that a passenger survived given that they are female (P(Survived | Female))

Conditional Probability

Probability that a passenger survived given that they are female (P(Survived | Female))

This uses **Conditional Probability**:

$$P(Survived|Female) = \frac{P(Female \cap Survived)}{P(Female)}$$

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

Conditional Probability

This uses **Conditional Probability**:

$$P(Survived|Female) = \frac{P(Female \cap Survived)}{P(Female)}$$

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

$$P(Survived|Female) = \frac{P(Female \cap Survived)}{P(Female)} = \frac{233}{314} \approx 0.7417$$

Conditional Probability

Probability that a passenger survived given that they are male (P(Male | Female))

Gender	Did Not Survive (0)	Survived (1)	Total
Female	81	233	314
Male	468	109	577
Total	549	342	891

Conditional Probability

Probability that a passenger survived given that they are male (P(Male | Female))

$$P(Survived|Male) = \frac{109}{577} \approx 0.1889$$

Interpretation: Given the passenger is male, there is only an **18.89% chance** he survived.

Baye's Theorem

Bayes' Theorem is a way to update our beliefs about the probability of an event based on new evidence.

In simple terms:
It tells us how likely something is, *given* that we already know something else has happened.

Baye's theorem

Formula of Bayes' Theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:

- $P(A|B)$: **Posterior probability** (Probability of A *given* B has occurred)
- $P(B|A)$: **Likelihood** (Probability of B *given* A is true)
- $P(A)$: **Prior probability** (Initial belief about A)
- $P(B)$: **Marginal probability** (Total probability of B happening)

Baye's theorem example

What's the probability a survivor was female?

That's $P(Female \mid Survived)$

We use **Bayes' Theorem**:

$$P(Female|Survived) = \frac{P(Survived|Female) \cdot P(Female)}{P(Survived)}$$

$$= \frac{0.742 \cdot 0.352}{0.384} \approx \frac{0.261}{0.384} \approx 0.682$$

> So around **68.2% of the survivors were women**. Bayes for the win!

Difference

Feature	Conditional Probability	Bayes' Theorem
Definition	Probability of A given B	Updates probability of A after B is observed
Known Values	(P(A and B)), (P(B))	(P(B))
Purpose	Direct computation	Reversing or updating beliefs
Focus	Descriptive	Inference / learning from data
Use Case	"If it's raining, what's the chance the ground is wet?"	"The ground is wet — what's the chance it rained?"

Multiple Choice Questions (MCQs)

Q1: If events A and B are independent, $P(A \cap B)$ is:
A) $P(A) + P(B)$
B) $P(A) – P(B)$
C) $P(A) \times P(B)$
D) None of the above

Q2: What is the sample space for a coin toss?
A) {H, T}
B) {1, 2}
C) {Red, Black}

Multiple Choice Questions (MCQs)

Q1: If events A and B are independent, P(A ∩ B) is:
A) P(A) + P(B)
B) P(A) – P(B)
C) P(A) × P(B) ✅
D) None of the above

Q2: What is the sample space for a coin toss?
A) {H, T}
B) {1, 2}
C) {Red, Black}
Answer: A) {H, T}

Session Summary

In this session, we have discussed about:

- **Define** probability, sample space, and events.
- Understand key properties of probability.
- **Calculate** probabilities for dice, coins, and cards **with step-by-step solutions**.

Reference Links

- 10 Must-Know Probability Concepts for Every Aspiring Statistician and Data Scientist
- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- 10 Statistical Concepts Every Aspiring Data Scientist Should Know — The Davangers' Guide!
- Top 20 Interview Questions on Descriptive and Inferential Statistics
- 10 Must-Know Statistics Interview Questions (With Real Examples & Formulas!)